

BiLSTM vs Feedforward Neural Networks for Toxicity Detection

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Abstract

Automated detection of toxic online content is a critical task for fostering safer and more inclusive digital communities. This work formulates toxicity detection as a multi-label classification problem and compares two neural architectures on the Jigsaw Toxic Comment Classification dataset: a baseline Feedforward Neural Network (FFNN) that applies a per-token projection and mean-pooling over word embeddings, and a more advanced Bidirectional LSTM (BiLSTM) with an attention mechanism that captures sequential context and weights informative tokens. Both models are trained with Binary Cross-Entropy with Logits and positive class weighting to mitigate severe label imbalance, with predicted probabilities thresholded at 0.3 to compensate for the rare-label imbalance. Across six toxicity categories, the BiLSTM with attention outperforms the FFNN on subset accuracy (0.8269 vs. 0.7828), macro precision (0.2614 vs. 0.2394), and macro F1 score (0.3665 vs. 0.3514), while the FFNN retains a slight edge in macro recall (0.8884 vs. 0.8674). Per-sample agreement analysis on 153,164 test comments reveals strong per-label correlation between the two models ($r \in [0.747, 0.787]$), confirming that they capture broadly similar toxicity signals but disagree most on the highest-frequency *toxic* label. Rare classes such as *threat*, *severe_toxic*, and *identity_hate* remain difficult for both architectures due to extreme class imbalance. The results show that sequential modeling with attention yields more selective, higher-precision toxicity predictions than a simple averaging baseline.

1 Introduction

Toxic online interactions often contain language such as insults, threats, and hate speech, which can lead to individuals feeling harassed and being socially excluded. For this reason, social media platforms aim to develop fair, transparent, and accurate models that contribute to safer online spaces and more inclusive communities. Because moderating content manually is very expensive and can even introduce individual biases from each moderator, these platforms have increasingly relied on automated tools that can detect toxicity and abusive content. This project focuses

on building two distinct, but similar models that classify user comments into six toxicity categories using the Jigsaw Toxic Comment Classification dataset [2]. The goal of this project is to design, train, and evaluate a baseline FeedForward Neural Network (FFNN) against an improved advanced Bidirectional LSTM (BiLSTM) with attention as a higher proportion of samples are judged toxic in the presence of contextual information [7], and then compare their performance using standard multi-label evaluation metrics.

Toxicity classification is a multi-label problem where each comment may exhibit zero, one, or multiple toxic behaviors. Early research and traditional approaches relied on surface-level features, e.g., logistic regression using bag-of-words approaches [9, 10], but even though these approaches were reported to be highly predictive [8] and easily interpretable, they still suffered with false positives as the presence of certain patterns could lead to misclassification [6], which is a problem I also encountered with my models and will elaborate upon later. More recent advances in Natural Language Processing (NLP) highlight the effectiveness of deep models such as Convolutional Neural Networks (CNN) [4], Long Short-Term Memory Networks (LSTM) [1] which is well-suited for sequential text, and transformer-based architectures (like BERT) [3] which learns contextual word embeddings instead of static embeddings and have all shown a clear advantage in toxicity detection [11, 12].

Unlike traditional binary sentiment or toxicity tasks, I frame the task as a multi-label classification problem since the labels are not mutually exclusive, where each comment may belong to zero, one, or several toxicity categories simultaneously (e.g., *toxic*, *severe_toxic*, *obscene*, *threat*, *insult*, and *identity_hate*). Each label is treated as an independent binary classification task, but all are trained jointly via a shared neural network, which is why this task requires models that can capture overlapping patterns of abusive language and predict multiple outputs at once. Unlike softmax-based single-label tasks, this setup uses sigmoid activation per label and Binary Cross-Entropy with Logits (BCEWithLogitsLoss).

2 Methodology

This project’s methodology consists of four stages: pre-processing, feature representation, model training, and evaluation. Each comment that trains and tests the models undergoes the following preprocessing steps:

- Lowercasing every character to maintain consistency in the vocabulary.
- Removing punctuation and special characters (which may also remove valuable cues like emojis, so further discussed below).
- Splitting comments into individual words based on whitespace for tokenization.
- Truncating comments that are longer than 75 tokens and padding shorter comments with <PAD> so the models can all operate on fixed-sized input tensors, which covers the majority of cases while keeping model size and training time manageable.
- Limiting the vocabulary to the 20,000 most frequently used words in the training set (replacing any other words with a special <UNK> token).

Each token is mapped to an integer index, which is then passed through a trainable embedding layer (for this project, the dimension is set at 100) that learns a dense vector representation for each word, allowing the model to learn meaningful semantic structures. Both models are then trained using the following configuration:

- BCEWithLogitsLoss, which is suitable for independent multi-label outputs.
- Positive class weighting (pos_weight), which comes from the frequency of each label and works to counter heavy class imbalance.
- The Adam optimizer with a learning rate of $\eta = 0.001$ and weight decay of 0.001 for regularization.
- Gradient clipping at L2 norm 1.0 at each step to prevent exploding gradients.
- Mini-batch training of size 64, which balances performance and efficiency.
- 25 training epochs.

The datasets used in this project originate from the publicly available Jigsaw Toxic Comment Classification Challenge hosted on Kaggle, which contains comments scraped from Wikipedia discussion pages, specifically from “administration pages where editors can discuss improvements to articles or other Wikipedia pages” [5], where multiple human raters manually annotate each comment.

A summary of the shared and model-specific hyperparameters used to train both models is given in Table 1.

Table 1: Hyperparameters used for the Baseline FFNN and the Advanced BiLSTM with Attention.

Hyperparameter	FFNN	BiLSTM
Vocabulary size	20,000	20,000
Max. sequence length	75	75
Embedding dimension	100	100
Hidden size	64	128
Hidden layers	2	2
Dropout	—	0.4
Output units	6	6
Optimizer	Adam	Adam
Learning rate	0.001	0.001
Batch size	64	64
Epochs	25	25
Loss function	BCE w/ Logits	BCE w/ Logits

2.1 Baseline FFNN

The baseline FFNN treats each comment as a bag of token-level projections. Its architecture consists of:

1. Converting token indices to embeddings.
2. Applying a per-token fully connected projection of size 64 with a Rectified Linear Unit (ReLU) activation, which transforms each token’s embedding independently before aggregation.
3. Mean-pooling the projected token vectors across the sequence, which removes word order while keeping the model simple and computationally efficient.
4. Passing through a second fully connected hidden layer of size 64 with ReLU activation.
5. An output layer of 6 units (one per toxicity label).

The ReLU activation function used in the hidden layers is defined as $\text{ReLU}(x) = \max(0, x)$, while the sigmoid activation applied to the output logits is defined as:

$$\sigma(x) = \frac{1}{1 + e^{-x}}.$$

2.2 Advanced BiLSTM with Attention

The advanced BiLSTM captures sequential patterns and long-distance dependencies, and is augmented with an attention mechanism that learns to weight informative tokens within each comment. Its architecture consists of:

1. Converting token indices to embeddings.
2. Two stacked BiLSTM layers of size 128 each (which output a forward and backward sequence representation of $\text{hidden_size} \times 2$), with inter-layer dropout of 0.4 applied between the stacked layers.

3. An attention layer that computes a context vector as a weighted sum of the final BiLSTM hidden states, where the weights are produced by a learned linear scoring function over each hidden state followed by a softmax normalization.
4. A fully connected layer of size 128 with ReLU activation and dropout 0.4 for regularization.
5. An output layer of 6 units.

Concretely, given the BiLSTM hidden states $\{h_1, h_2, \dots, h_T\}$ at each time step, the attention mechanism computes a score $e_t = w^\top h_t + b$ for each token, normalizes the scores with a softmax to produce attention weights $\alpha_t = \frac{\exp(e_t)}{\sum_{k=1}^T \exp(e_k)}$, and combines the hidden states into a context vector $c = \sum_{t=1}^T \alpha_t h_t$ that is then passed through the hidden FC layer and the output layer. The learnable parameters w and b allow the model to highlight tokens like slurs or threat phrasing that disproportionately drive toxicity, rather than averaging their contribution uniformly across the sequence. A side-by-side comparison of the two architectures is shown in Figure 1.

The Binary Cross-Entropy with Logits loss used to jointly train both models is given by:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^6 [y_{ij} \log \sigma(z_{ij}) + (1 - y_{ij}) \log(1 - \sigma(z_{ij}))],$$

where N is the batch size, $y_{ij} \in \{0, 1\}$ is the ground-truth label for sample i and class j , and z_{ij} is the corresponding output logit.

At evaluation time, predicted sigmoid probabilities are thresholded at 0.3 rather than the conventional 0.5 to counteract the strong class imbalance, which otherwise causes the model to predict all-zeros for the rare labels. All precision, recall, F1, and confusion-matrix values reported in the Results section use this 0.3 threshold.

3 Results

Before training even began, to understand the distribution and structure of the comment texts, some basic statistics of the training dataset were computed: the median comment length is 35 tokens, the mean length is around 65 tokens, the 75th percentile is located at 73 tokens, the 85th percentile is located at 111 tokens, and the longest comment length was 1,403 tokens long. These results show that the dataset is heavily skewed toward shorter comments, with a long tail of very long inputs, which is why I decided to keep the maximum length of tokens at 75 as a reasonable compromise between computational efficiency and retention of meaningful content.

Table 2: Per-Label Metrics for the Baseline FFNN.

Label	Precision	Recall	F1 Score
toxic	0.3958	0.8984	0.5495
severe_toxic	0.1509	0.9375	0.2600
obscene	0.4117	0.9192	0.5687
threat	0.0214	0.8462	0.0417
insult	0.3610	0.9122	0.5173
identity_hate	0.0956	0.8169	0.1711

Table 3: Per-Label Metrics for the Advanced BiLSTM.

Label	Precision	Recall	F1 Score
toxic	0.4908	0.8529	0.6230
severe_toxic	0.1382	0.9625	0.2418
obscene	0.4350	0.9216	0.5910
threat	0.0191	0.7692	0.0373
insult	0.4020	0.8951	0.5548
identity_hate	0.0832	0.8028	0.1508

3.1 Per-Label Metrics

The standard multi-label evaluation metrics used above are computed as follows, where TP, FP, and FN denote true positives, false positives, and false negatives, respectively:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}},$$

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

Figure 2 visualizes the per-label correlation between the two models alongside the mean absolute difference of their predicted probabilities. The two models agree most strongly on rare labels with little signal to capture (*threat*, $r = 0.787$) and disagree most on *toxic*, the highest-frequency label, where the BiLSTM’s sequential modeling diverges most noticeably from the FFNN’s bag-of-embeddings approach.

3.2 Per-Label Confusion Matrices

The per-label confusion matrix counts for both models are summarized in Tables 4 and 5, where TP, FP, FN, and TN denote true positives, false positives, false negatives, and true negatives, respectively.

3.3 Comparison of Models

Some key points to note with each model. With the baseline FFNN, there was strong recall across all labels (often greater than 0.90), there was low precision, especially

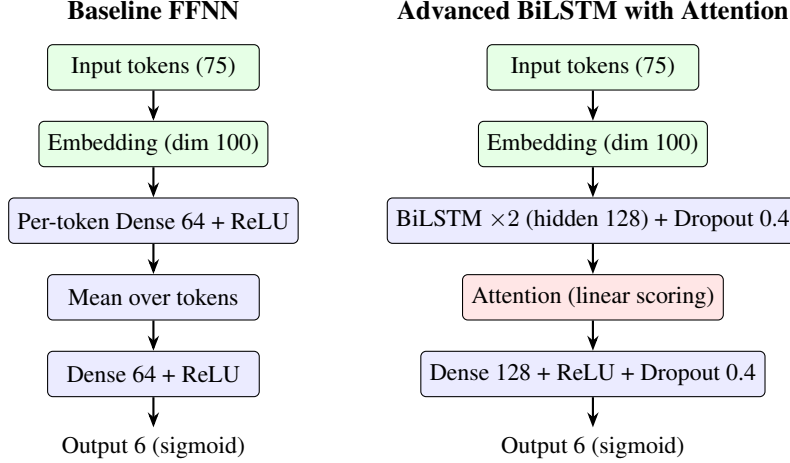


Figure 1: Side-by-side architectures of the Baseline FFNN (left) and the Advanced BiLSTM with Attention (right). The FFNN applies a per-token linear projection before mean-pooling, then a single hidden FC layer, discarding word order. The BiLSTM processes the sequence in both directions through two stacked layers, learns token weights through a linear attention scoring function, and passes the resulting context vector through a hidden FC layer before classification.

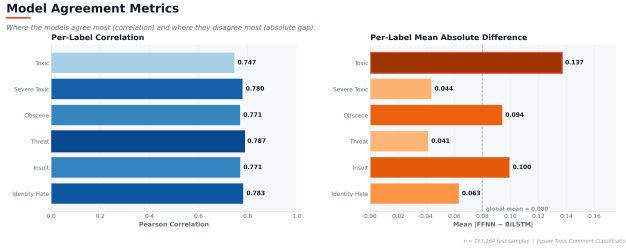


Figure 2: Per-label Pearson correlation and mean absolute difference between the FFNN and BiLSTM predicted probabilities across the six toxicity categories. Correlation is consistently high (≥ 0.747), but the absolute prediction gap is largest for the high-frequency *toxic* label.

for rare labels such as *threat* and *identity_hate*, and its F1 scores ranged from ~ 0.04 (*threat*) to ~ 0.57 (*obscene*), showing how the baseline model tends to over-predict toxic labels (high recall, low precision), which is common with highly imbalanced multi-label tasks like this. With the advanced BiLSTM, F1 scores increased across *toxic*, *insult*, *obscene*, and *identity_hate*, and it produced fewer false positives compared to the simpler baseline, basically outperforming the baseline model across almost all meaningful metrics (e.g., macro precision/accuracy/F1 score), demonstrating its strong ability to detect toxic comments while maintaining high recall, likely as a result of its sequential understanding of text. However, rare classes such as *threat*, *identity_hate*, and *severe_toxic* still remain challenging to correctly classify due to extreme label imbalance.

The average predicted probability per label, shown in

Table 4: Per-Label Confusion Matrix Counts for the Baseline FFNN.

Label	TP	FP	FN	TN
toxic	672	1026	76	6205
severe_toxic	75	422	5	7477
obscene	387	553	34	7005
threat	11	503	2	7463
insult	374	662	36	6907
identity_hate	58	549	13	7359

Figure 3, gives further insight into model behavior. The FFNN assigns slightly higher mean probabilities to the more common positive-leaning labels (*toxic*, *insult*, *threat*), which is consistent with its tendency to over-predict, while the BiLSTM is more confident on *severe_toxic* and *identity_hate*, suggesting it differentiates these rarer categories more strongly. The signed mean prediction gap in Figure 4 confirms this asymmetry: the FFNN’s confidence advantage on *toxic* (+0.0111) is the single largest gap, while the BiLSTM dominates on *severe_toxic* (−0.0076).

3.4 Prediction Agreement Analysis

To better understand how the two models differ at the sample level beyond aggregate metrics, the predicted probabilities of both models were compared across the entire test set of 153,164 comments. Figure 5 shows the prediction probability distributions for each label, where both models concentrate mass near zero, consistent with the heavy

Table 5: Per-Label Confusion Matrix Counts for the Advanced BiLSTM.

Label	TP	FP	FN	TN
toxic	638	662	110	6569
severe_toxic	77	480	3	7419
obscene	388	504	33	7054
threat	10	513	3	7453
insult	367	546	43	7023
identity_hate	57	628	14	7280

Average Model Confidence per Label

Sorted by FFNN mean. Δ shows the gap between models.

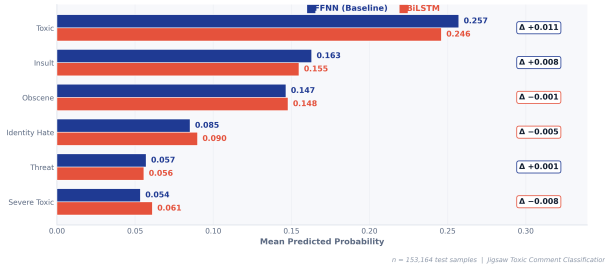


Figure 3: Average predicted probability per label for the FFNN and BiLSTM. The Δ annotations show the per-label gap between the two models.

class imbalance, but the BiLSTM produces a sharper, more peaked distribution near zero on the high-frequency labels (*toxic*, *insult*, *obscene*), suggesting it is more decisive in rejecting non-toxic comments. The FFNN by contrast spreads more probability mass into the mid-range, which is consistent with its higher false positive rate.

Figure 6 shows the per-sample distribution of prediction gaps (FFNN – BiLSTM) on a log scale. Most samples fall near zero, but the long, roughly symmetric tails on either side indicate that the two models do disagree substantially on a non-trivial fraction of comments, even when their mean predictions are similar.

Finally, Figure 7 presents per-sample hexbin density plots of the BiLSTM probabilities against the FFNN probabilities for each label. The dark concentration along the bottom-left of each plot reflects the dominant non-toxic class, and the Pearson correlations ranging from $r = 0.747$ (*toxic*) to $r = 0.787$ (*threat*) quantify the strong but imperfect linear relationship between the two models’ outputs.

4 Discussions

The BiLSTM with attention clearly outperformed the baseline FFNN model in nearly all aspects, proving that better sequential modeling and understanding increase precision. A bidirectional context helps a model capture nuance

Signed Mean Prediction Gap

Which model is, on average, more confident for each toxicity label.

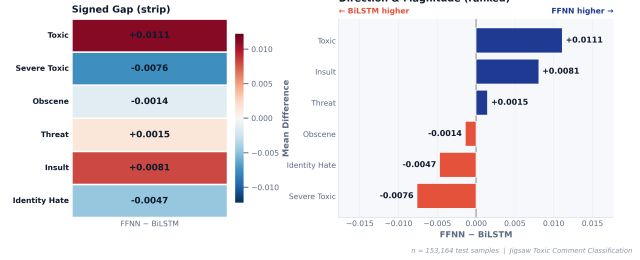


Figure 4: Signed mean prediction gap (FFNN – BiLSTM) per label. Positive values (blue) indicate labels where the FFNN is more confident on average; negative values (red) indicate labels where the BiLSTM is more confident.

like sarcasm, multi-word insults, and certain threat and hate phrasing, and attention helps highlight toxic keywords like insults or slurs. The BiLSTM strikes a better balance than the FFNN, producing higher quality, more selective toxic predictions, making it significantly more effective for automating moderation, toxicity monitoring, and content filtering, where minimizing the number of false alarms and missed toxic content is necessary for healthier online discussions.

Some unexpected observations that came with the project include how certain labels, such as *threat* and *identity_hate*, remained difficult for both models to correctly classify, mainly as a result of their low frequency and how threats can be expressed and interpreted in many various ways. Also, during BiLSTM’s training period, its average training loss for each epoch was fluctuating/oscillating (between ~ 0.42 and ~ 0.39), suggesting that the model was potentially highly sensitive to the learning rate, that it was overfitting on minority labels, and that it was being influenced by the “long-tail” inputs (comments greater than 150 tokens). I was also pleasantly surprised by how well the basic FFNN performed against the BiLSTM with attention, especially in recall, which can be an indicator that these shallow networks can still detect basic toxicity, possibly because toxicity can appear as obvious words or phrases.

4.1 Limitations

Limitations of this project were not nonexistent, as I performed my training on a smaller subset of the training dataset (39,892 samples) rather than the complete Jigsaw training dataset (160,000 samples), mainly because of my laptop’s computational constraints, including limited GPU availability. This may have caused the models to not have captured the full diversity of toxic language present in the complete dataset, explaining the dataset imbalance that I encountered, which led to a high false positive rate, unstable

Table 6: Comparison of overall metrics between the Baseline FFNN and the Advanced BiLSTM.

Metric	Baseline FFNN	Advanced BiLSTM	Better Model
Subset Accuracy	0.7828	0.8269	BiLSTM
Macro Precision	0.2394	0.2614	BiLSTM
Macro Recall	0.8884	0.8674	Baseline
Macro F1 Score	0.3514	0.3665	BiLSTM

F1 values, and the models overfitting to the majority labels (of which `pos_weight` partially helped but not fully solved). Other limitations that were affected by my computational constraints include:

- Using a fixed vocabulary size of 20,000 words, so rare slurs or coded language were often ignored.
- Setting the maximum sequence length to 75 tokens/words, so long context for lengthy comments or multi-sentence toxicity were disregarded.
- Keeping the number of training epochs low, so the model would be cut short during its training.

I also didn’t include pretrained embeddings, opting towards randomly initialized embeddings, which lack inherent semantic relationships (so “father” and “mother” would not be inherently closer in the embedding space than “father” and “princess”) and prior knowledge (models learn effective representations from scratch).

4.2 Potential Improvements

Potential improvements to this area of study include:

- Integrating pretrained embeddings, like GloVe, into both models, as this will result in a big improvement in their performance, especially with precision, rare label detection, and their F1 score by better understanding uncommon, nuanced, and subtle toxic slurs.
- A further dive into fine-tuning different parameters and hyperparameters like the learning rate, batch size, LSTM hidden size, number of layers, dropout rate, and the embedding dimension could further enhance stability and performance (I briefly and manually tried a few different options, which is why I landed on different dropout and `hidden_size` numbers for the FFNN and BiLSTM).
- Comparing the performance results of both these models against a modern benchmark transformer model like DistilBERT will be intriguing to see just how much sequence encoders with more sophisticated attention mechanisms outperform traditional RNN-based architectures.
- Improving the preprocessing stage by implementing lemmatization, spelling correction, and more robust

text-cleaning techniques like emoji normalization and handling elongated words (“soooo”) could also improve the model’s performance.

- Incorporating the entire training and test dataset will enable the model to learn from a more balanced dataset, resulting in more stable, reliable performance estimates that are necessary for real-world applications.

5 Conclusion

This project compared a baseline Feedforward Neural Network against a Bidirectional LSTM with attention on the multi-label task of classifying toxic online comments from the Jigsaw Toxic Comment Classification dataset. The BiLSTM with attention outperformed the FFNN on subset accuracy, macro precision, and macro F1 score, while the FFNN held a slight edge in macro recall, reflecting its tendency to over-predict toxic labels at the cost of precision. Per-sample agreement analysis on 153,164 test comments showed strong but imperfect Pearson correlation ($r \in [0.747, 0.787]$) between the two models, indicating that they capture broadly similar toxicity signals but diverge most on the highest-frequency *toxic* label, where sequential context and attention add the most value. Both architectures struggled on the rarest categories (*threat*, *severe_toxic*, *identity_hate*), confirming that extreme class imbalance remains the dominant challenge in this task. Overall, the results support the intuition that sequential modeling with attention produces more selective, higher-precision toxicity predictions than a simple averaging baseline, while also suggesting that shallow networks remain surprisingly competitive on the recall side. Future work integrating pretrained embeddings, transformer-based encoders such as DistilBERT, and more aggressive handling of rare-label imbalance should narrow the remaining gaps, particularly on the long tail of identity- and threat-related categories.

6 Code Availability

The Python code of the analysis of this work can be found on GitHub: <https://github.com/EduardoRebollar/BiLSTM-vs-Feedforward-Neural-Netwo>

Prediction Probability Distributions

How confidently each model labels comments across the six toxicity categories.

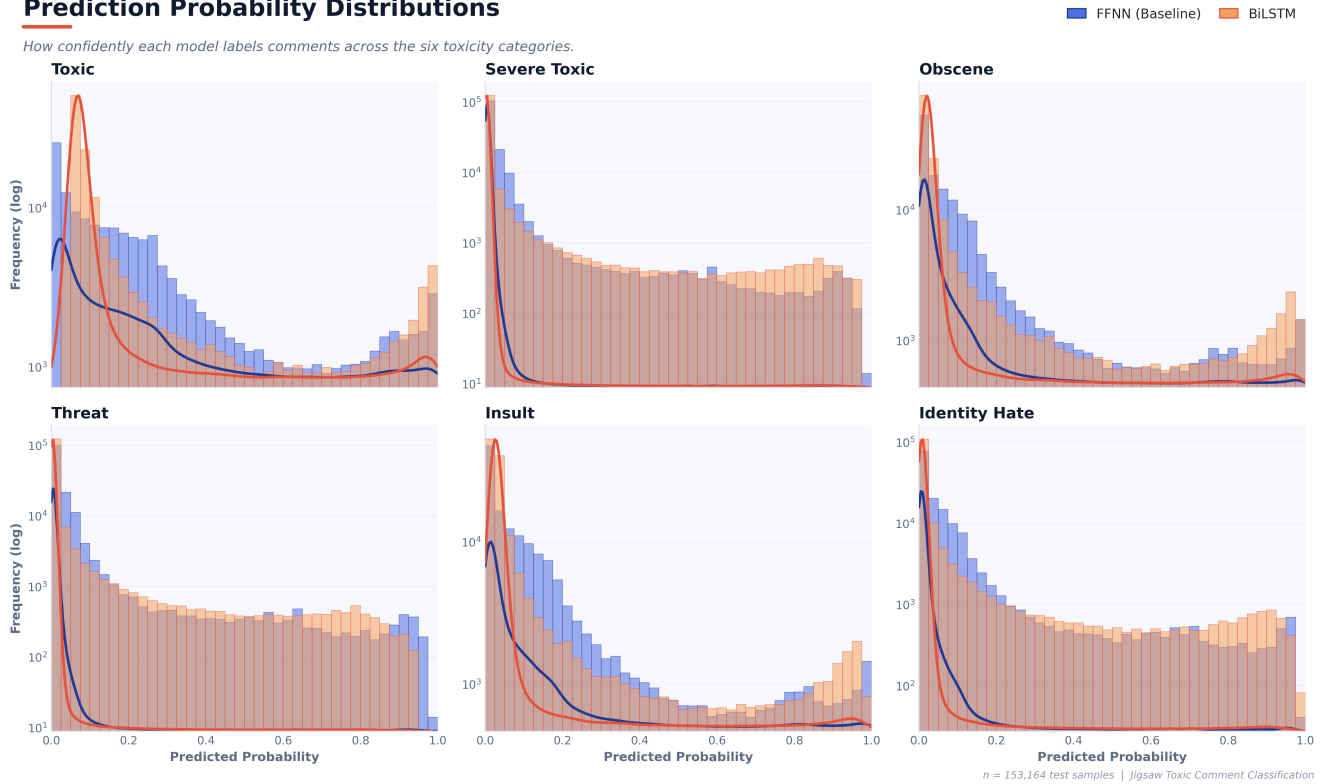


Figure 5: Per-label distributions of predicted probabilities for the FFNN (blue) and BiLSTM (orange) on the test set. The BiLSTM produces sharper peaks near zero on the high-frequency labels, indicating more decisive rejection of non-toxic comments.

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Prediction Differences (FFNN – BiLSTM)

Per-sample gap distributions. Blue tint $\Rightarrow$ FFNN higher; coral tint $\Rightarrow$ BiLSTM higher.

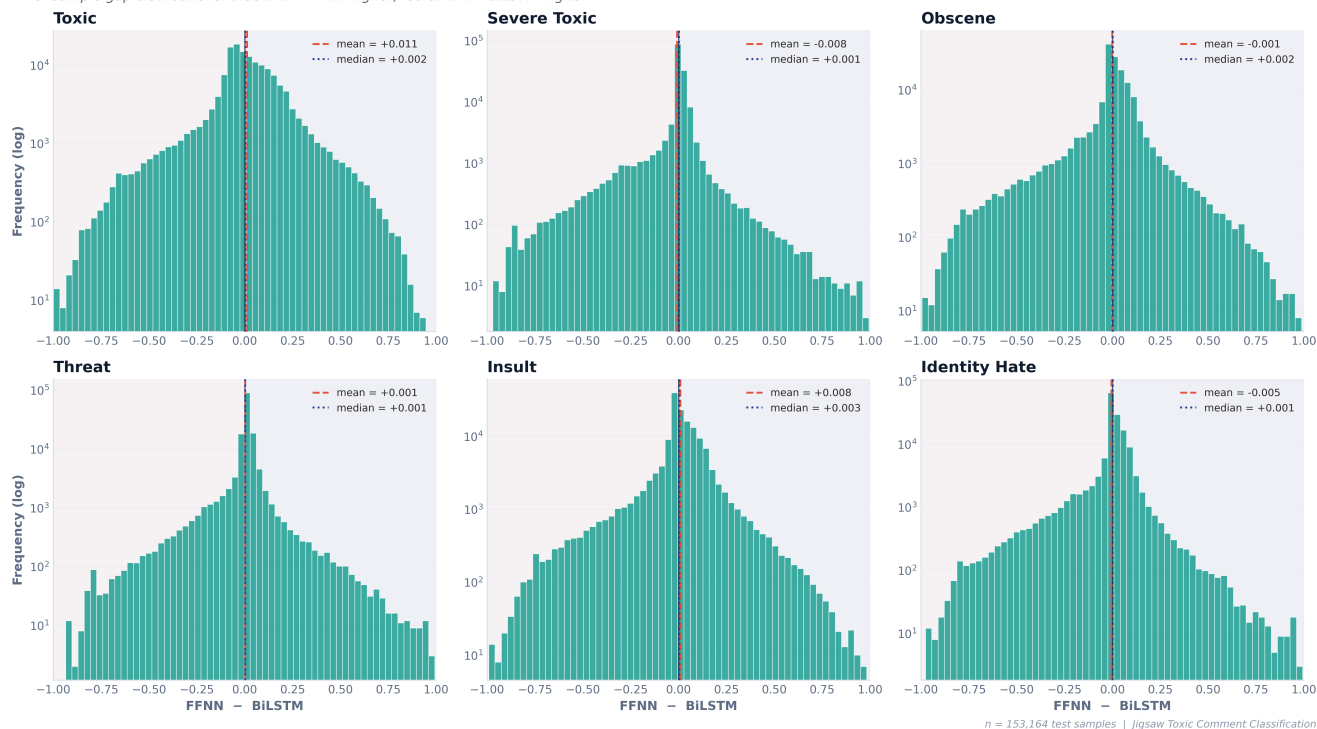


Figure 6: Per-sample distribution of prediction gaps (FFNN – BiLSTM) for each label on a log-frequency scale. The long symmetric tails indicate that the two models disagree substantially on a non-trivial subset of samples, even when their mean predictions are close.

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Per-Sample Agreement

Hexbin density of FFNN vs BiLSTM predictions. Diagonal = perfect agreement.

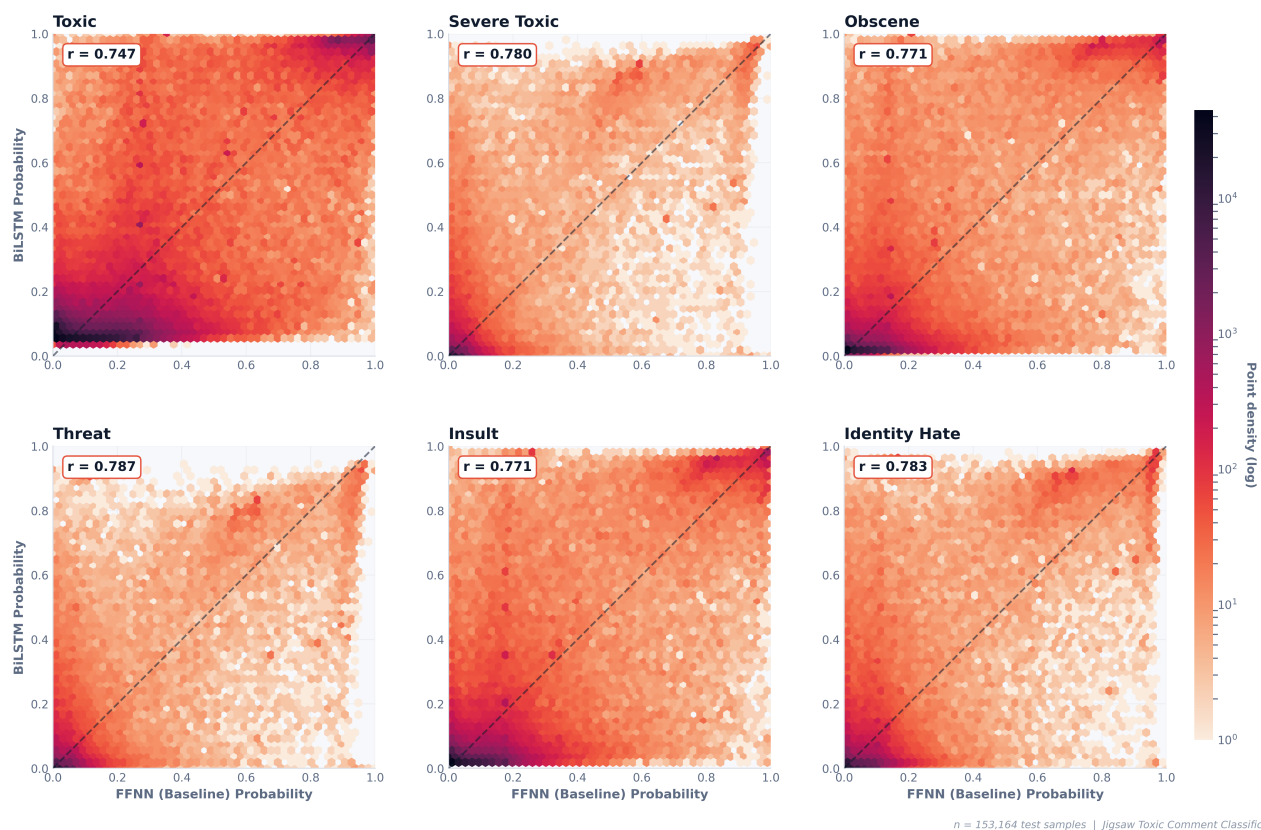


Figure 7: Per-label hexbin density plots of BiLSTM probabilities against FFNN probabilities on the test set. The dashed diagonal marks perfect agreement; Pearson correlations are shown in each panel.